

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

Course Code: EET204
Course Name: ELECTROMAGNETIC THEORY

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

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|----|--|---|
| 1 | Explain divergence of a vector field with its physical significance | 3 |
| 2 | Evaluate the gradient of the scalar field $\frac{3z}{\rho} \cos \phi$ | 3 |
| 3 | Sketch equipotential surfaces for a positive test charge | 3 |
| 4 | Illustrate the conservative nature of electrostatic field | 3 |
| 5 | Using Ampere's circuital law, obtain the expression for magnetic field intensity due an infinite wire carrying current | 3 |
| 6 | What do you mean by magnetic vector potential | 3 |
| 7 | Explain the significance of Poynting vector | 3 |
| 8 | What is skin effect and explain its significance | 3 |
| 9 | List the reasons for Electromagnetic Interference | 3 |
| 10 | Explain impedance matching in transmission lines | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|---|
| 11 | a) State and prove Stokes' theorem | 7 |
| | b) Evaluate the curl of the following vector fields a) $\mathbf{P} = xy \mathbf{a}_x + y^2 \mathbf{a}_y + xz \mathbf{a}_z$ | 7 |
| | b) $\mathbf{Q} = \rho z^2 \mathbf{a}_\rho + \rho \sin^2 \phi \mathbf{a}_\phi + 2\rho z \sin^2 \phi \mathbf{a}_z$ | |
| 12 | a) Explain spherical to rectangular coordinate system transformation | 7 |
| | b) Express the vector $\mathbf{A} = \rho \sin \phi \mathbf{a}_\rho + \rho \cos \phi \mathbf{a}_\phi - 2z \mathbf{a}_z$ in rectangular coordinates | 7 |

Module -2

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|----|---|---|
| 13 | a) State and explain Gauss's law | 7 |
| | b) Using Gauss's law, determine the electric field intensity due to an infinite sheet charge of uniform surface charge density. | 7 |

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- 14 a) Derive the expression for capacitance of a coaxial cable 7
b) Obtain Poisson's and Laplace's equations for a homogeneous material 7

Module -3

- 15 a) State and explain Biot Savart's law 7
b) Obtain the expression for magnetic field intensity on the axis of a rectangular loop carrying current 7
- 16 a) Explain the boundary conditions for electric field in a dielectric-dielectric interface 7
b) State and explain the continuity equation for current 7

Module -4

- 17 a) Derive wave equations from Maxwell's equations 7
b) Explain propagation of uniform plane waves in a lossy dielectric medium 7
- 18 a) Define the following terms: phase velocity, group velocity, Attenuation constant and Propagation Constant 6
b) State and explain Poynting theorem 8

Module -5

- 19 a) Obtain the transmission line equations and its solution 10
b) State the transmission line parameters 4
- 20 a) Obtain the expression for propagation constant and characteristic impedance in the case of a lossless transmission line 7
b) A low loss coaxial cable of characteristic impedance $50\ \Omega$ is terminated by a resistive load of $150\ \Omega$. Calculate Voltage standing wave ratio. If the maximum voltage in SWR is 30V, find the minimum voltage 7
